AI Hallucinations in Legal Cases

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Research Question, Exploratory Analysis, and Decision Tree

Research Question

Artificial intelligence tools are increasingly being used in legal settings, but AI-generated hallucinations can introduce inaccurate information into legal documents and proceedings. Our research question is:

What characteristics of legal cases involving AI hallucinations are associated with whether a professional sanction is issued?

We focus on characteristics of the parties involved and the primary legal field of the case. Our response variable is whether the case resulted in a professional sanction.

Data Preparation

```
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   4.0.2     v tibble    3.2.1
v lubridate 1.9.3     v tidyr     1.3.1
v purrr     1.0.4
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco
```

```
library(rpart)
library(rpart.plot)
```

```
Warning: package 'rpart.plot' was built under R version 4.4.3
```

```
hallucination <- read_csv("hallucination_cases.csv")
```

```
Rows: 2078 Columns: 18
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (17): Case Name, Court, State(s), Party(ies), AI Tool, Hallucination It...
date  (1): Date

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
hallucination <- hallucination |>
  mutate(
    sanction = factor(
      `Professional Sanction`,
      levels = c("No", "Yes")
    ),

    # Group party types for exploratory analysis
    party_group = case_when(
      `Party(ies)` == "Lawyer" ~ "Lawyer",
      `Party(ies)` == "Pro Se Litigant" ~ "Pro Se Litigant",
      TRUE ~ "Other"
    ),

    # Variables used in the decision tree
    party = factor(`Party(ies)`),
    legal_field = factor(`Legal Field Primary`),
    alleged = factor(Alleged),
    vendor_disputed = factor(`Vendor Disputed`)
  ) |>
  drop_na(sanction)
```

##Exploratory Analysis##

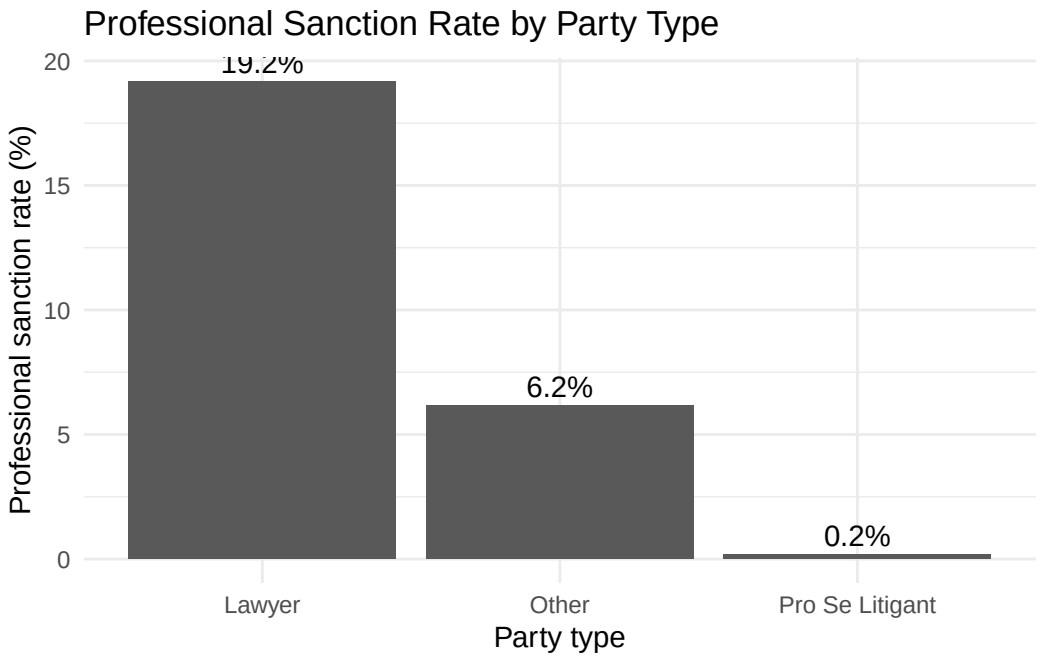
Because our response variable is whether a professional sanction was issued, we first examine whether the sanction rate differs across party types.

```r
party_summary <- hallucination |>
  group_by(party_group) |>
  summarise(
    cases = n(),
    sanctions = sum(sanction == "Yes"),
    sanction_rate = mean(sanction == "Yes") * 100
  )

party_summary
```

```
# A tibble: 3 x 4
  party_group     cases sanctions sanction_rate
  <chr>           <int>     <int>         <dbl>
1 Lawyer            819       157        19.2
2 Other              65         4         6.15
3 Pro Se Litigant  1194         2         0.168
```

```r
ggplot(
  party_summary,
  aes(x = party_group, y = sanction_rate)
) +
  geom_col() +
  geom_text(
    aes(label = paste0(round(sanction_rate, 1), "%")),
    vjust = -0.4
  ) +
  labs(
    title = "Professional Sanction Rate by Party Type",
    x = "Party type",
    y = "Professional sanction rate (%)"
  ) +
  theme_minimal()
```



The exploratory analysis shows substantial differences in professional sanction rates across party types. Cases involving lawyers have a considerably higher sanction rate than cases involving pro se litigants, while cases classified as other party types fall between these groups. This suggests that party type may be an important predictor of professional sanctions and motivates its inclusion in the decision tree.

##Decision Tree##

To examine whether party type and other case characteristics jointly predict professional sanctions, we fit a classification tree using party type, primary legal field, alleged misconduct, and whether the AI vendor disputed the hallucination.

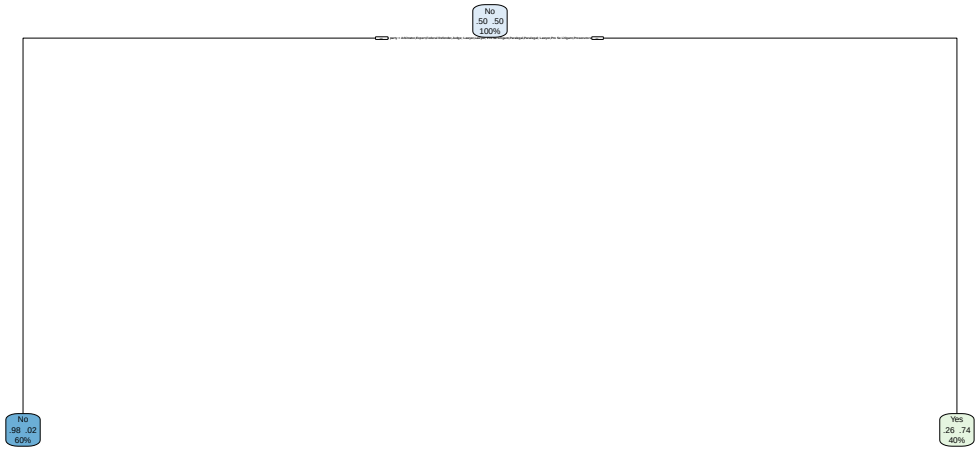
Because professional sanctions are relatively uncommon in the dataset, we use equal class priors so that the tree does not simply favor the majority class of cases with no sanction.

```
#splitting and cleaning up data
train_id <- sample(1:nrow(hallucination), round(0.8 * nrow(hallucination)))
hallucination_train <- hallucination[train_id,] |>
  select(sanction, party, legal_field, alleged, vendor_disputed)
hallucination_test <- hallucination[-train_id,] |>
  select(sanction, party, legal_field, alleged, vendor_disputed)
```

```
tree <- rpart(
  sanction ~ party + legal_field + alleged + vendor_disputed,
```

```
  data = hallucination_train,
  method = "class",
  parms = list(prior = c(0.5, 0.5)),
  control = rpart.control(
    cp = 0.001,
    minsplit = 20,
    minbucket = 10
  )
)
```

```
rpart.plot(
  tree,
  type = 2,
  extra = 104,
  fallen.leaves = TRUE,
  faclen = 0,
  varlen = 0,
  split.cex = 0.35,
  cex = 0.7,
  compress = FALSE
)
```




```
rpart.rules(
  tree,
```

```
  style = "tall",
  cover = TRUE
)
```

```
sanction is 0.02 with cover 60% when
    party is Arbitrator or Expert or Federal Defender or Judge; Lawyer or Lawyer; Pro Se Liti

sanction is 0.74 with cover 40% when
    party is Judge or Lawyer or Lawyer; Judge
```

The decision tree identifies party type and legal field as important predictors of professional sanctions. The first split is based on party type, indicating that this variable provides the greatest reduction in classification error among the predictors considered. One branch contains approximately 60% of the cases and has a very low model-assigned probability of a sanction.

Among the remaining cases, legal field provides an additional important split. The tree identifies combinations of party type and legal field associated with substantially higher predicted probabilities of professional sanctions. In particular, one terminal node has a model-assigned probability of approximately 83% for a professional sanction and contains approximately 8% of the cases.

Overall, the tree suggests that professional sanctions are not evenly distributed across cases involving AI hallucinations. Instead, the predicted likelihood of a sanction varies substantially depending on the characteristics of the parties involved and the legal field of the case.

Random Forest

```
library(randomForest)
```

```
randomForest 4.7-1.2

Type rfNews() to see new features/changes/bug fixes.


Attaching package: 'randomForest'

The following object is masked from 'package:dplyr':

    combine
```

```
The following object is masked from 'package:ggplot2':

    margin
```

```
#creating random forest
hallucination_train <- na.omit(hallucination_train)
forest <- randomForest(sanction ~ party + legal_field + alleged + vendor_disputed,
                       data = hallucination_train,
                       ntree = 500,
                       importance = TRUE,
                       proximity = TRUE)

forest
```

```
Call:
 randomForest(formula = sanction ~ party + legal_field + alleged +      vendor_disputed, data
               Type of random forest: classification
                     Number of trees: 500
No. of variables tried at each split: 2

        OOB estimate of  error rate: 7.64%
Confusion matrix:
      No Yes class.error
No  1341   3 0.002232143
Yes  108   0 1.000000000
```

Model Comparison

```
tree_predict <- predict(tree, newdata = hallucination_test, type = "class")
confusion_tree <- table(hallucination_test$sanction, tree_predict)
confusion_tree #accuracy = 0.65625, sensitivity = 0.947
```

```
     tree_predict
       No Yes
  No  241 147
  Yes   1  27
```

```
hallucination_test <- na.omit(hallucination_test)
forest_pred <- predict(forest, newdata = hallucination_test)
confusion_forest <- table(hallucination_test$sanction, forest_pred)
confusion_forest
```

```
     forest_pred
       No Yes
  No  332   4
  Yes  24   2
```

```
((350)/(350+2+25))
```

```
[1] 0.928382
```

```
#classification error
#tree
mean(tree_predict != hallucination_test$sanction) #= 0.3125
```

```
Warning in `!=.default`(tree_predict, hallucination_test$sanction): longer
object length is not a multiple of shorter object length
```

```
Warning in is.na(e1) | is.na(e2): longer object length is not a multiple of
shorter object length
```

```
[1] 0.4399038
```

```
#forest
mean(forest_pred != hallucination_test$sanction) #= 0.071
```

```
[1] 0.07734807
```

The accuracy of the single decision tree is 0.656, while the accuracy of the forest is 0.928. The sensitivity of the tree is 0.947 and the sensitivy of the forest is 0. We see an interesting result from comparing the two models in that despite its higher accuracy, the random forest resulted in a lot of false positives. This could be because the original data is very skewed in that the there are more `no sanctions` than `sanctions`.

Conclusions

There were some limitations to this analysis as the data isn't clean. There were missing rows that had to be cleaned up for the random forest, and the response is very skewed in the actual data as described above. Nevertheless, we choose to compare the models by their misclassification error rates. The decision tree's misclassification error rate is 0.3125, while the misclassification error rte of the random forest is only 0.071. With a lower error rate, we decide the random forest as the better model.